Towards capital efficient margining systems

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Abstract

This paper evaluates if correlations between stocks can be used to design more capital efficient margins which do not compromise on protection. It focuses its attention on the cash market in India and uses very basic measures of correlations i.e. covariance matrices to evaluate the trade-offs between a gross-based and portfolio-based margining system.

The paper finds that gross margins are about 1.5 times the portfolio margins on an average. The portfolio-based margining system does well in stable periods. However, in times of crisis, such as the financial crisis of 2008, it is not able to provide protection, while the gross-based margin does. The paper thus sets the framework for studying more robust models that can perform well even in crisis periods.

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1 Introduction

This paper evaluates if correlations between stocks can be used to design more capital efficient margins which do not compromise on protection. It focuses its attention on the cash market in India and uses very basic measures of correlations i.e. covariance matrices to address three questions: What would be the margin required if a portfolio-based margining system were used? How much lower would this be than the gross-based margin posted by the exchange? How much higher is the risk in the portfolio-based system as compared to the gross-based system?

The paper finds that gross margins are about 1.5 times the portfolio margins on an average. The portfolio based margining system does well in stable periods. However, in times of crisis, such as the financial crisis of 2008, it is not able to provide protection, while the gross-based margin does. However, the estimators that were used in this paper are static estimators and do not take into account dynamic models of forecasting correlations.

It is also important to note that the comparison in this paper is between the portfolio-based margin and the VaR margin posted by the exchange. If the exchange levies additional margins (such as the initial margin, mark-to-market margin, special margin etc.) on top of the VaR margin, the portfolio-based margin may well be enough to offer protection in times of crisis, even if by itself it is not.

The capital efficiency and protection trade-off of even the most basic static models provide a motivation for studying more robust models that can perform well even in crisis periods.

Section 2 lays out the problem of defining margining systems that are required to be efficient both from a systemic risk management perspective as well as from a capital efficiency perspective. The key to solving this is to use portfolio marginning systems that take as input covariance estimators of returns that are described in Section 2.1. However, the crux of the problem is the testing framework used to select from among multiple choices, which is described in Section 2.2. The data used for the analysis and the first stage of results is presented in Section 3. Section 4 presents a comparison of the best performing portfolio based margins against the performance of the margins that are presently used by the equity exchanges, both in periods of crisis (where there is greater stress on the marginning systems) and non-crisis. A strategy for future research is chalked out in 5.

2 Marginning systems for clearing corporations

An important component of the modern financial system is the clearing corporation, which acts as a *legal counterparty* to every trade, and simultaneously assume the role of a bank and an insurance company (Bernanke, 1990). These are typically referred to as the *Central Clearing CounterParty* (CCCP)

CCCPs manage their legal liability by levying *margins* on each trade that members enter into. These are collected upfront and become the first line of defence against a future situation in which a member defaults on a trade obligation. These are called the *initial margins*. In addition, CCCPs monitor the positions taken by members in their effort to ensure that market participants are able to honour their commitments. These are the *mark-to-market* (MTM) margins. Together, these margins become collateral to the trades done by every member, on both sides of the trade.

Prudential margin design deals with the question of how much is the optimal margin setting that will provide adequate protection against default, and has been the subject of much research (Baer, France, and Moser, 1994). The basic trade-off in margin design is that between high costs to the trading member and prudential concerns about systemic risk management.

Conservative margins ensure that the CCCPs will be able to ensure settlement of trades, using margins and collections to make good trades that are not honoured. However, from the traders perspective, high margins imply that a lot of capital gets locked up, increasing the cost to the trader. At the extreme, excessive margins may inhibit trades and have an adverse impact on the liquidity of the security. This increase in liquidity risk raises the cost to the CCCP in both lower clarity in risk measurement as well as if defaulting positions have to be unwound.

On the other hand, margins that are too low could be inadequate as protection in times of systemic stress, such as the financial crisis of 2008. An optimal design of CCCP margins is therefore of paramount importance.

A feature of the existing margin system is that the margin for a portfolio of different stocks is set on a *gross* basis. This implies that a margin is calculated for each stock in the members portfolio, and this stock-level margin is added to obtain the total margin on the portfolio. There are no offsets that are considered for stocks that may be correlated to each other, where the member can have taken positions that are of an opposite view.

Gross marginning systems with zero offsets for correlations can therefore, end up being very expensive with respect to portfolios where the risk of default is actually a lot lower than if the margins could be collected on a *net* basis, including offsets for opposite exposures. Bernanke (1990) discusses the role of the *offset* in determining how far a clearing corporations guarantee exists. Under a net margining system, it is conceivable that the goal of the clearing house is to only protect members from the defaults of other members, and not to guarantee all futures contracts. This is an important distinction to make, as it implies that margins need be much lower than they are today.

The CCCP and the regulator for the equity markets in India have taken extremely conservative positions so far, and demand very high margins for all trades on the cash as well as the derivatives markets. By regulatory mandate, there are three types of margins: Value at Risk (VaR) or intial margin, MTM margin and Extreme Loss margin (ELM).[1] In this system, there is no offset allowed across different underlyings.[2] Given this conservative position, any suggested alternative to replace the current system of marginning needs to have a basis not only in computationally efficient models but also those that can be demonstrated to perform according to the risk parameters set by the CCCP/regulator.

The key models in calculating net margins for a portfolio of underlyings are those for robust estimation of the covariance of their returns. Earlier efforts assumed a symmetric distribution like the multivariate normal and focussed on estimating the covariance matrix accurately. Today, two paths of development of these models involve:

- Dynamic estimators of the covariance matrix such as the various multivarite GARCH estimators, or
- Joint tail behaviour of the returns distribution without much development of the remainder of the distribution, like Copulas.

What is almost as important as the models themselves, though, is the testing strategy to establish the performance of these estimators in a back-testing framework. In addition to regulatory compulsions like the Basle II Amendments in 2005 and 2006, which emphasised back-testing of risk measurement models that are used in calculating capital allocation, there is a focus on testing the model for *coherence* (Alexander, 2009). In this paper, we attempt

[1] A brief explanation of these is provided in the Appendix.

[2] Some offset is allowed for different securities on the same underlying. For example, a long position on a stock, and a short position on a 2-month futures contract on the same stock, can be offset.

to take a step in this direction by comparing the behaviour of alternative risk measures in periods of stress (like the 2008 global credit crisis) and the periods of calm.

We next proceed to presenting the alternative covariance matrix estimators, followed by the testing framework we use.

2.1 Variance-covariance matrix estimators

Covariance matrix estimation has been the subject of much research for its importance in portfolio optimization (Markowitz, 1952) and risk management. The success of both are conditional upon accurate estimation of the moments of the returns distribution, among other things.

The most traditional method of using the sample covariance matrix is problematic because it leads to a huge estimation error. This is because the number of parameters to be estimated are far greater than the number of observations. Also, if N the number of stocks is greater than T, the length of the time series of returns available for stocks, the sample covariance matrix is estimated with a lot of error. In particular, the estimator may exhibit large off-diagonal elements (Paffka, Potters, and Kondor, 2004). Inverting such a matrix amplifies the error and the matrix may not be invertible at all (Michaud, 1989).

There are three broad methods that can be used to resolve this problem:

1. Factor models
2. Shrinkage estimators
3. Estimators using intra-day data

In this paper, we focus on the first two models. We use the following methods for calculating correlations. We note that these are static models of estimation. Dynamic models such as Garch and MvGarch models can also be used to forecast volatility, and will be considered in future research.

Historical covariance matrix

This is the simple sample covariance matrix between stock i and j for $n-1$ time periods (for example)

$$cov(i,j) = \frac{1}{N-1}\sum_{t=1}^{n}(r_{it} - \bar{r_i})(r_{jt} - \bar{r_j})$$

r_{it} is the return of stock i in period t, and $\bar{r_i}$ is the mean return of stock i.

Single Index Market Model

The Single Index Market Model (SIMM) flows from the Arbitrage Pricing Theory which assumes that in a large economy returns of any asset Y_i over the risk free rate are correlated with excessive returns of K factors. The simplest such model is the single-index model of (Sharpe, 1963), which assumes that the only relevant factor is the market index.

The SIMM model thus involves regressing stock i's returns against the market

$$x_{it} = \alpha_i + \beta_i x_{0t} + \epsilon_{it}$$

Let b_i denote the slope estimate, d_{ii} the residual variance estimate, $D =$ diagonal matrix containing residual variance estimates d_{ii} and $s_{00}^2 =$ sample variance of market returns. The covariance matrix is then computed as

$$F = s_{00}^2 bb' + D$$

This model is interesting because if the market return can capture the cross-sectional risks, the number of parameters to be estimated for the covariance matrix gets reduced.

Vasicek beta corrected

This is the same as the SIMM method, except that the β's from the regression are adjusted using the Vasicek method. Least squares provides the best unbiased estimator in the sense that the expected value of the estimator is supposed to equal the true value. However, as pointed out by Vasicek (1973), the reverse is actually true. We know the sample value of the parameter and on this basis we infer the distribution of the parameter. He therefore suggests a Bayesian approach to the calculation of the parameters.

In this model, the posterior distribution of beta is assumed to be normal with mean b'' and variance $s_b''^2$.

$$b'' = \frac{\frac{b'}{s_b'^2} + \frac{b}{s_b^2}}{\frac{1}{s_b'^2} + \frac{1}{s_b^2}}$$

$$s_b''^2 = \frac{1}{\frac{1}{s_b'^2} + \frac{1}{s_b^2}}$$

Here, s_b^2 is the variance of beta; b' and s_b' are parameters obtained from the prior distribution. b' is typically the average value of beta from the population of stocks from which the particular stock comes from. The values so obtained are then used compute the covariance matrix using the same formula as the SIMM.

Vasicek beta corrected, Garch (1,1) nifty variance

The models discussed so far assume homoskeasticity of the error term. However, it may be that the variance of the error term is not equal, and in the presence of such heteroskedasticity, the standard errors and confidence intervals may be too narrow, thus comprising on precision. Garch models are typically used in such problems: these models compute the variance of each error term, and thus correct for heteroskedasticty (Engle, 2001).

We use the Garch (1,1) estimator to model variance of the market return s_{00}^2 and use the same in the Vasicek beta model. The Garch specification asserts that the best predictor of the variance in the next period is a weighted average of the long-run average variance, the variance predicted for this period, and the new information in this period that is captured by the most recent squared residual.

The Garch (1,1) specification is as follows:

$$s_{t+1}^2 = \omega + \alpha s_t \epsilon_t^2 + \beta s_t$$

Here, α, β and $1 - \alpha - \beta$ are the constants, and $\sqrt{\frac{\omega}{1-\alpha-\beta}}$ is the long run average variance. $\alpha + \beta < 1$. The updation rule thus only requires knowing the previous forecast s and the residual ϵ.

Shrinkage

The central idea in these models is to use some sort of a 'shrinkage' technique that will reduce some of the estimation error that occurs in the sample covariance matrix. A shrinkage estimator is typically an optimally weighted average of the sample matrix with an invertible covariance matrix estimator (Disatnik and Benninga, 2006). This optimal weighting is supposed to ensure that it diversifies the errors made by each of the component estimator and these errors cancel out (Bengtsson and Holst, 2002; Jagannathan and Ma, 2003).

Ledoit and Wolf (2003) propose a shrinkage estimator that is a weighted average of the single-index covariance matrix and the sample covariance matrix. They develop a formula for the optimal shrinkage intensity and find that it is able to select portfolios of stocks with low out-of-sample variance compared to existing estimators.

We use the method developed by Opgen-Rhein and Strimmer (2007) to compute the covariance matrix. The shrinkage target in this method is as follows

$$t_{ij} = \begin{cases} s_{ii} & \text{if } i = j \\ 0 & \text{if } i \neq \text{j} \end{cases}$$

The shrinkage intensity is calculated as follows:

$$\hat{\lambda}^* = \frac{\sum_{i \neq j} \widehat{Var(s_{ij})}}{\sum_{i \neq j} s_{ij}^2}$$

2.2 Testing framework

The objective of the paper is to evaluate various covariance estimators from the viewpoint of their performance on the twin objective of maximising capital efficiency while achieving the stated level of safety required by the firm or the regulator. The latter is based on statistical tests comparing model outcomes against realised values, typically in an out-of-sample setting. The former objective of optimal capital efficiency will have a basis in the utility function of the application of the margins. In the context of VaR models used for margining systems, the literature includes work such as Lopez (1999).

In this paper, we use both aspects in a sequential manner to rank different covariance estimators. Since the base application of the covariance estimators is as an input into portfolio margin calculation, we base the testing framework in the context of portfolio VaR. Each estimator is used to calculate the VaR of a given portfolio. Then, the comparison of these estimators is done as follows:

1. Test for the statistical efficiency of the VaR estimators in an out-of-sample setting. This involves testing the VaR values forecasted by each estimator and comparing them against the realised portfolio returns. The difference between the two should satisfy the properties defined by the VaR parameters, such as correct unconditional and conditional coverage, independence of outcomes, etc.

   These tests would help eliminate those estimators that do not satisfy satistical efficiency, and ideally, identify a subset from the original set of selected covariance estimators.

2. Once a subset of estimators has been selected, these would be tested in the context of economic efficiency, which would identify which estimator enables the best use of capital. At the end of this stage, we would be left holding those estimators which satisfy both conditions of statistical and economic efficiency.

In this paper, we focus primarily on the first phase of selection which is the selection of the covariance estimator that is statistically efficient in the context of portfolio (net) marginning. For this, we use two broad statistical testing strategies. The strategies involve computing the one-step ahead forecast from each of the covariance matrix measures and testing for an equally weighted long-long portfolio.

Regression based strategy

We use the strategy suggested by (Engle and Sheppard, 2008). The portfolio variance, denoted by $h^P_{t|t-1}$ is predicted using the one-step ahead forecast. We also compute the square of the realized portfolio return and denote it by $(r^P_t)^2$. The forecast standardized residuals are formed as follows

$$sr_t = \frac{(r^P_t)^2}{h^P_{t|t-1}}$$

A joint test that the forecast is unconditionally correct and not auto-correlated is carried out using a regression as follows

$$sr_t = \mu + \rho sr_{t-1} + v_t$$

The null hypothesis to be tested is $H_0 : \mu = 1, \rho = 0$. Since the null can be rejected by either cause, separate tests for the two elements of the null are conducted. The regression also controls for possible standardized residual heteroskedasticity by using the White robust standard errors.

The intuition behind this framework is that the constant captures the failure of the specification to correctly forecast unconditional portfolio variance. The lag of the standardized portfolio return is supposed to capture the neglected dynamics.

Coverage statistics

We use a Value-at-Risk (VaR) based framework to test the five models. A VaR forecast by definition should be such that a 95% VaR should be exceeded 5% of the time, and that the probability of the VaR being exceeded at $t+1$ should remain 5% after conditioning for all information at time t. This is tested using the "conditional coverage" framework developed by (Christoffersen, 1998). The idea behind conditional coverage is that a correctly specified VaR should generate the pre specified failure rate conditionally at every point in time.

We use the three step testing framework which includes testing for correct unconditional coverage (UC), independence (I) and for correct conditional coverage (CC).[3] The null for the UC test is the failure probability p against the alternative hypothesis that the failure probability is different from p. The null for I is that failure process is independent against the alternative hypothesis of a first-order Markov process. The LR statistic for both these tests is asymptotically distributed as $\chi^2(1)$ under the null hypothesis. The CC test is done by testing the null-hypothesis of a first-order Markov failure process with a different transition probability matrix (TPM). It is essentially a joint test of unconditional coverage and independence. The conditional coverage LR statistic is the sum of the unconditional coverage LR statistic and the independence LR statistic, and asymptotically distributed as a $\chi^2(2)$ under the joint null hypothesis.

The procedure is well explained by (Sarma, Thomas, and Shah, 2003) and is as follows: One-step ahead VaR forecasts v_t estimated at the 99% level of significance are computed and compared with the realizations of the returns

[3]This testing procedure has also been used by (Sarma, Thomas, and Shah, 2003; McAleer and Veiga, 2008)

series r_t. A new series is I_t is defined as follows

$$I_t = \begin{cases} 1 & \text{if } r_t < v_t \\ 0 & 0 \text{ otherwise} \end{cases}$$

The I_t series, also known as the failure process is tested for unconditional coverage, independence and conditional coverage.

3 Data and results of the statistical tests

We use close-price data of stocks in the Nifty and Nifty junior indices. Returns are computed as log-differences of daily close prices. Thus $R_{it} = log(P_{it}/P_{i,t-1})$, where P_{it} is the close price of stock i at time t.

We use data from 1 January 2001 to 31 January 2011. This gives us a total of 2517 days. We use a rolling window estimation of 1500 days. This means that we use data from the first 1500 days to estimate the various covariance matrices and form a forecast for day 1501. The 1501st day corresponds to 21 December 2006. We then use data from day 2 to day 1501 to obtain a forecast for day 1502. We move through time to generate one-step ahead out-of-sample forecasts. We have a total of 1016 one-step ahead forecasts.

We randomly pick 5 stocks from the set of 100, and form an equally-weighted portfolio of these stocks. This is an example of a long-long portfolio. We intend to form a long-short portfolio in future work. The five stocks randomly chosen include Reliance, Hindalco, Federal Bank, Ashok Leyland, and HCL Tech. The plots of the returns of the five stocks and Nifty for the given period are given in Figure 1.

We observe that for each of the stock and Nifty, the period between 2008 and 2009 reflects a high volatility. The descriptive statistics for the series are given in Table 1.

Except for the Hindalco series, all series have similar means and medians, close to zero. The minimum varies between -13.05 of Nifty to -21.75 of Hindalco. The maximum varies between 15.38 of HCL Tech and 19.36 of Reliance. There is a wide variation between the skewness of all the series, the kurtosis is roughly similar. The high value of the kurtosis indicates that there may be extreme observations. The Jarque-Bera test strongly rejects the null of normally distributed returns.

Figure 1 Returns of the stocks in the portfolio

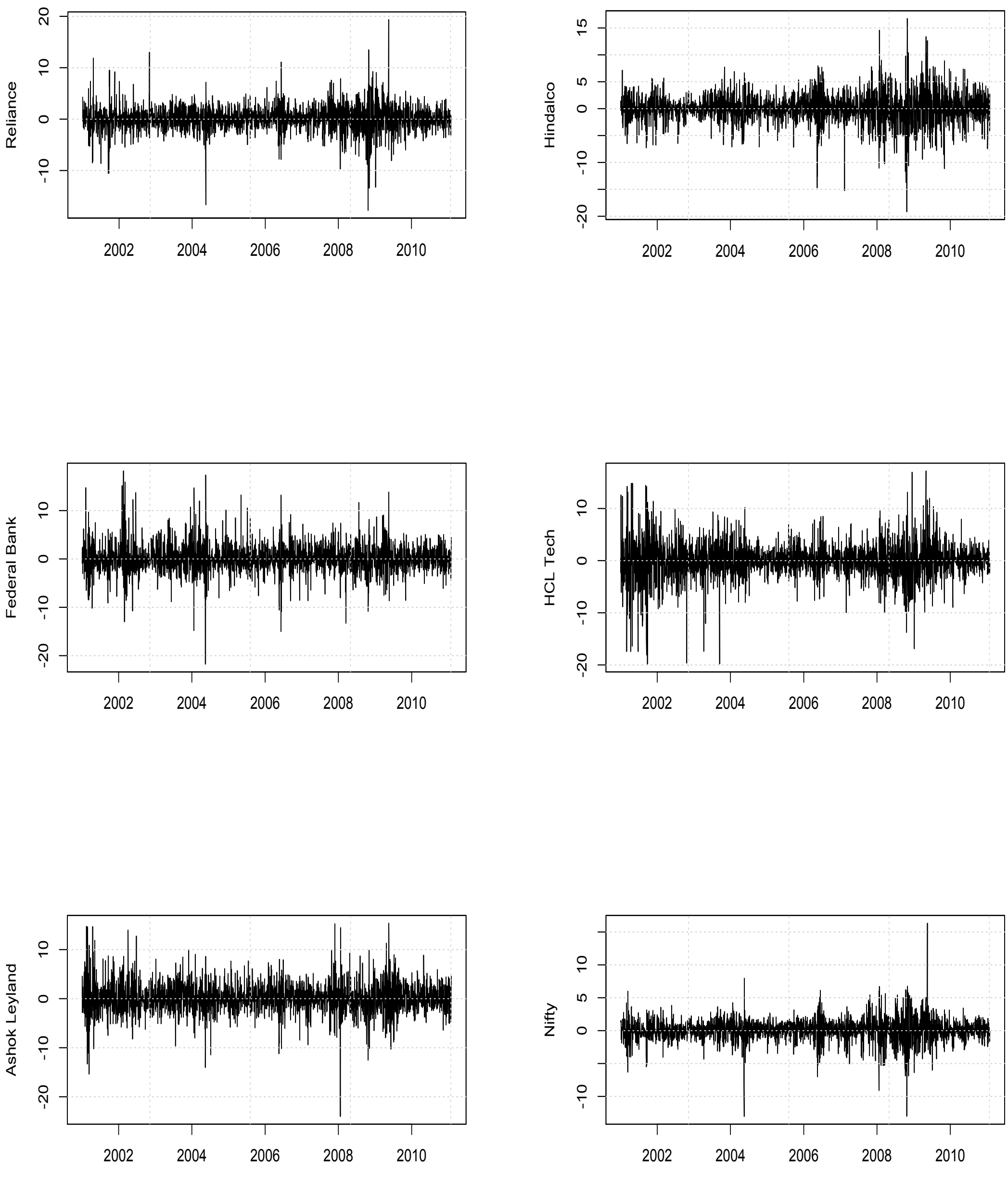


Table 1 Descriptive statistics for the returns

Statistic	Reliance	Hindalco	Federal Bank	HCL Tech	Ashok Leyland	Nifty
Mean	0.07	0.05	0.13	0.02	0.10	0.05
Median	0.12	0.07	-0.03	0.01	0.00	0.13
Maximum	19.36	16.75	18.22	17.21	15.38	16.33
Minimum	-17.76	-19.17	-21.75	-19.85	-24.01	-13.05
SD	2.39	2.76	3.00	3.49	3.03	1.68
Skewness	-0.16	-0.26	0.14	-0.39	0.07	-0.29
Kurtosis	9.65	7.39	8.99	7.83	7.43	11.33
Jarque-Bera	4658.6	2056.75	3771.55	2514.03	2066.21	7313.24

3.1 Regression based strategy

The results are presented in Table 2.

Table 2 Regression based tests

Covariance matrix	$\mu = 1$	$\rho = 0$	$\mu = 1$ & $\rho = 0$	R^2
Historical	1.02	0.25*	5.36*	0.06
	(0.12)	(0.03)		
SIMM, historical variance	1.002	0.25*	4.97*	0.06
	(0.12)	(0.03)		
VB, historical variance	1.001	0.25*	4.94*	0.06
	(0.12)	(0.03)		
VB, nifty with Garch(1,1)	0.99	0.03	0.34	0.001
	(0.08)	(0.03)		
Shrinkage	1.01	0.25*	5.29*	0.06
	(0.12)	(0.03)		

We find that except for the Vasicek model with Garch (1,1) nifty variance, all the other models fail the test for $\rho = 0$ and the joint test for $\mu = 1 \& \rho = 0$. The Vasicek beta model with a Garch (1,1) nifty variance appears as the strongest candidate amongst the five models.

3.2 Unconditional coverage, conditional coverage and independence

Table 3 presents the results of the three LR tests. Out of the five models, four violate all the three tests. The only model that performs well on all three tests is the Vasicek beta model, with nifty forecasted using a Garch (1,1) process.

Table 3 Coverage tests

Covariance matrix	UC	Ind	CC
Historical	7.56*	8.72*	16.43*
SIMM, historical variance	6.87*	7.02*	14.04*
VB, historical variance	6.87*	7.02*	14.04*
VB, Garch(1,1) nifty variance	0.01	3.26	3.46
Shrinkage	6.86*	7.02*	14.03

3.3 Robustness check

It could be argued that the results obtained in the previous section are just a function of the one random portfolio that is tested. As a robustness check, we therefore ran the same models for 1000 random portfolios for the same time period as discussed above. In Table 4, we present the proportion of models that were rejected by the various models by both the tests.

Table 4 Proportion of portfolios rejected by both models

	Coverage tests			Regression based
	UC	Ind.	CC	$\mu = 1$ & $\rho = 0$
Historical	0.9	0.80	0.99	0.79
SIMM, historical variance	0.83	0.80	0.98	0.84
VB, historical variance	0.83	0.80	0.99	0.84
VB, nifty with Garch(1,1)	1	0.09	0.99	0.27
Shrinkage	0.88	0.80	0.99	0.80

We find that both the tests show that the models do not generally perform very well on predicting forecasts.[4] There is also quite a bit of variation in the two tests: while the VB, Garch (1,1) model does well on the regression based strategy and the independence test, it does not do so well on the unconditional coverage test.

Considering these results, we proceed to testing the capital efficiency of the models using the VB, nifty with Garch (1,1) model. We test the validity of the five models again for the relevant out-of-sample periods to ensure that we have chosen the correct specification.

[4]This is similar to the results found by (Engle and Sheppard, 2008) when they tested the regression based strategy on other dynamic models.

4 Comparing margins in crisis vs. calm periods

A part of the objective of the paper is to evaluate how much lower the cost would be if portfolio margining can be done instead of gross margining, while keeping to similar levels of safety. Towards this objective, we now proceed to calculate the difference in the gross margin that is currently calculated and levied in the equity markets, and the net margin from a portfolio based system.

In order to emulate a *stress-test* scenario, we include the performance of the two margin systems in a period of crisis, and juxtapose the statistical and economic performance in this period in comparison to a relatively less volatile period in the market. We therefore focus on two distinct time periods:

- In the first time period the out-of sample period corresponds to 16 January 2008 - 29 February 2008. This is the period of the "financial crisis", and therefore important to evaluate the performance of the portfolio-based VaR system.
- The second time period is where the out-of-sample period corresponds to 4 January 2011 - 30 January 2011 which is a relatively "stable period", after the financial crisis.

We create 1000 random portfolios and evaluate all the models for the two time periods for each portfolio. Table B.1 and Table C.1 in the appendix present the proportion of the models that were rejected by the coverage tests in the crisis period and the stable period. Consistent with results in section 3.2 the Vasicek beta model with Garch (1,1) nifty variance does the best. In Table B.2 and C.2 we present the proportion of the models rejected using the regression based testing approach. Again, the Vasicek model with Garch (1,1) variance is rejected the least number of times.

We then calculate the portfolio VaR (at 99%)[5] using the Vasicek beta, Garch (1,1) model for each date in the two time periods specified above. We compare the portfolio VaR to the VaR posted on the NSE website at the beginning of each of those dates.

NSE calculates VaR using the EWMA methodology:

$$\sigma_t = \sqrt{0.94 * \sigma_{t-1}^2 + 0.06 * r_t^2}$$

[5]The VaR estimate is defined as -2.33 * predicted volatility.

The margin rate is the higher of 3.5 times the volatility or 7.5% of value. In addition to this, there are several other layers of margins, but we abstract from those for this analysis as our focus is on understanding the difference between a *gross-based* system of calculating margins vs. a *portfolio-based* system. We assume that the extra layers of margins are the same in both instances.[6]

The VaR margin so posted on the NSE website for each stock is obtained and the total VaR is calculated by adding the VaR for each stock in the portfolio. For each date we evaluate the number of portfolios for which the actual portfolio return violated both the gross VaR limit and the portfolio VaR limit. This is done as follows:

$$B_{it} = \begin{cases} 1 & \text{if } r_{it} < v_{it} \\ 0 & 0 \text{ otherwise} \end{cases}$$

Here, t stands for the particular date in the out-of-sample period, and i stands for a portfolio. v_{it} stands for the portfolio VaR and gross VaR, depending on the VaR in question. For each t we then calculate the proportion of portfolios for which the actual portfolio return did indeed breach the VaR limit.

We find that in the crisis period, the portfolio VaR was breached on 8 of the 32 days by at least one of the 1000 portfolios, whereas gross VaR was violated only on 2 of the 32 days. In the stable period however, the portfolio VaR was violated only 4 out of the 19 days, while gross VaR was never violated.

In Table 5, we present the days as well as the proportion of portfolios for which both the portfolio VaR and the gross VaR were violated. We find that in the crisis period, particularly on the dates of 22 Jan 2008 and 25 Jan 2008, most of the portfolios breached the portfolio VaR threshold. This was not the case with the gross VaR thresholds.

While it is useful to know that the margins were (or were not) violated on a particular date, it does not answer the question of the cost at which this protection was achieved. We therefore present the ratio between the gross VaR and portfolio VaR (averaged across the 1000 portfolios) on each date. In stable times gross VaR has been about 1.50 times the portfolio VaR, while in the times of crisis, it was about 2.5 times the portfolio VaR.

In Figure 2 we present the difference between the margin posted by each

[6] A brief explanation of the margining system in India is presented in the Appendix.

Table 5 VaR comparison

Date	Proportion of portfolios for which Portfolio Return < gross VaR	Portfolio Return < portfolio VaR	Gross VaR/ portfolio VaR
Crisis period			
18 Jan 2008	0.00	0.26	0.86
21 Jan 2008	0.41	1.00	1.06
22 Jan 2008	0.004	0.88	1.92
23 Jan 2008	0.00	0.85	2.44
24 Jan 2008	0.00	0.18	2.79
25 Jan 2008	0.00	0.97	2.95
28 Jan 2008	0.00	0.01	2.94
29 Jan 2008	0.00	0.00	2.71
Post crisis stable period			
07 Jan 2011	0.00	0.07	1.51
08 Jan 2011	0.00	0.19	1.58
27 Jan 2011	0.00	0.05	1.48
31 Jan 2011	0.00	0.04	1.33

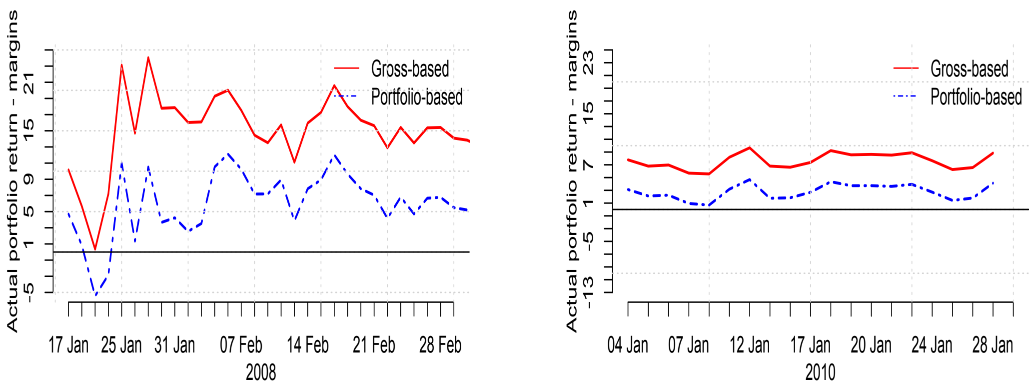
method and actual portfolio return observed on each date in the two time-periods.

We find that, in the months of January and February 2008, gross-based margins were far higher than the actual portfolio return observed, whereas the difference between the portfolio-based margins and the actual portfolio return was negative on certain days, as is also evident from Table 5. However, in the month of January 2010, the difference between the gross-based and portfolio-based margins was not as stark, even though gross-based margins continued to be more expensive than portfolio-based margins, with not much difference on the protection provided.

We also calculate the same statistic for the "pre-crisis" period (where the out-of-sample data corresponds to 21 December 2006 - 28 February 2007). We find that both the portfolio VaR and gross VaR thresholds are not crossed by the actual portfolio return. This shows that the gross based system of determining margins is extremely expensive, though it does prove protection against extreme events such as the financial crisis of 2008.

It is also important to note that the comparison here is between the portfolio-based margin to the VaR margin posted by the exchange. If the exchange levies additional margins (such as the initial margin, mark-to-market margin, special margin etc.) on top of the VaR margin, the portfolio-based margin

Figure 2 Difference between the VaR margins and the actual portfolio returns



may well be enough to offer protection in times of crisis.

5 Conclusion and future work

This paper sets the framework for evaluating the various margining schemes vis-a-vis their impact on protection and cost of capital. It finds that a portfolio VaR based on static covariance matrix measures is not able to protect in crisis times, such as the financial crisis of 2008. However, it is about 1.51 times less expensive than the gross margin during normal times.

The models used in this paper are very basic, and do not take into account dynamic forecasting techniques that can provide more robust forecasts of volatility. Future research will be directed towards two broad areas:

1. Building more sophisticated models to forecast volatility. These include:
   (a) Multivariate Garch models
   (b) Copula-based measures to model fat-tails
2. Developing a more robust testing framework. This includes:

(a) Applying the testing strategies discussed in the paper to long-short portfolios, and to portfolios with varying weights

(b) Applying portfolio management techniques to evaluate the various models

(c) Extending the 5 stock portfolio to 10 stock and 30 stock portfolios and to a portfolio consisting of futures and options.

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Appendix

A Margin systems in India

As part of the Comprehensive Risk Management System, BSE and NSE publish the margins requirement of every individual security at fixed time intervals. The calculation of margin requires the securities to be classified into three groups based on their liquidity and impact cost (reference) as presented in Table A.1. The daily margin payable by members consists of the following:

Table A.1 Classification of securities

Group	Trading Frequency (over the previous six months)	Impact Cost (over the previous six months)
Group I	At least 80% of the days	Less than or equal to 1%
Group II	At least 80% of the days	More than 1%
Group III	Less than 80% of the days	N/A

VaR margin
The Value at Risk (VaR) margin is a margin intended to cover the largest loss that can be encountered on 99% of the days (99% VaR). For liquid stocks, the margin covers one-day losses while for illiquid stocks, it covers three-day losses so as to allow the Exchange to liquidate the position over three days. This leads to a scaling factor of square root of three for illiquid stocks. For liquid stocks, the VaR margins are based only on the volatility of the stock while for other stocks, the volatility of the market index is also used in the computation. The VaR Margins are specified as follows for different groups of stocks:

Table A.2 VaR margins

Liquidity Categorization	One-Day VaR	Scaling factor for illiquidity	VaR Margin
Group I	Scrip VaR	1.00	Scrip VaR
Group II	Max(Scrip VaR, 3 x Index VaR)	1.73	Max(1.73 x Scrip VaR, 5.20 x Index VaR)
Group III	5 x Index VaR	1.73	8.66 x Index VaR

Extreme loss margin
The Extreme Loss Margin (ELM) for any security is higher of 5%, and 1.5 times the standard deviation of daily logarithmic returns of the stock price

in the last six months. This computation is done at the end of each month by taking the price data on a rolling basis for the past six months and the resulting value is applicable for the next month.

Special margin
Special margin may be imposed by exchanges from time to time on certain scrips as a surveillance measure and informed to the Members through notices.

Mark-to-market margin
The Mark-to-market (MTM) Margins are computed by marking each transaction to the closing price of the security involved at the end of trading. In case the security has not been traded on a particular day, the latest available closing price is considered as the closing price. In case the net outstanding position in any security is nil, the difference between the buy and sell values shall be considered as notional loss for the purpose of calculating the MTM margin payable.

B Out-sample: Jan 16, 2008 - Feb 29, 2008

Table B.1 Proportion rejected, 5 stock portfolio

Covariance matrix	UC	Ind	CC
Historical	0.79	0.12	0.85
SIMM, historical variance	0.83	0.09	0.86
VB, historical variance	0.83	0.09	0.86
VB, nifty variance forecasted using Garch(1,1)	0.02	0.46	0.64
Shrinkage	0.79	0.12	0.85

Table B.2 Proportion rejected, 5 stock portfolio

Covariance matrix	$\mu = 1$	$\beta = 0$	$\mu = 1$ & $\beta = 0$	$\bar{\mu}$	$\bar{\beta}$	
Historical	0.00	0.57	0.48	2.7	0.38	
SIMM, historical variance	0.00	0.57	0.51	2.85	0.38	
VB, historical variance	0.00	0.57	0.51		2.86	0.38
VB, nifty with Garch(1,1)	0.00	0.7	0.22	1.64	0.44	
Shrinkage	0.00	0.57	0.48	2.73	0.38	

C Out-sample: January 2011

Table C.1 Proportion rejected, 5 stock portfolio

Covariance matrix	UC	Ind	CC
Historical	0	0	0
SIMM, historical variance	0.002	0.00	0.002
VB, historical variance	0.002	0.00	0.002
VB, nifty variance forecasted using Garch(1,1)	0.09	0.01	0.1
Shrinkage	0.001	0.001	0.001

Table C.2 Proportion rejected, 5 stock portfolio

Covariance matrix	$\mu = 1$	$\beta = 0$	$\mu = 1$ & $\beta = 0$	$\bar{\mu}$	$\bar{\beta}$
Historical	0.54	0.021	0.58	0.54	0.08
SIMM, historical variance	0.48	0.021	0.53	0.56	0.08
VB, historical variance	0.48	0.021	0.53	0.56	0.08
VB, nifty with Garch(1,1)	0.03	0.03	0.03	1.06	0.1
Shrinkage	0.53	0.02	0.58	0.54	0.08